status word fault bit, such as free stop - fault state, according to a certain speed reduction stop - fault state, pause to lock the axis -The self reset is allowed to indicate whether the motor is allowed to automatically reset the fault bit of the current status word after the fault of the motor is eliminated, 1-allowed, 0-disallowed. The order of the corresponding bits is as follows:

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Malfunction Response Code [8]								Self-reset allowed [8]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Error code [16]															

Specific Definitions of Fault Response Codes

0- The state machine switches to the fault state (refer to "4.1 CiA402 state machine"), if $605E_h$ ($03C1_h$) >= 0, then the fault shutdown mode selected by $605E_h$ ($03C1_h$) is followed; otherwise: power tubes are fully switched off, free shutdown.

1- The state machine switches to the fault state (refer to "4.1 CiA402 state machine"), if $605E_h$ ($03C1_h$) >= 0, then follow the fault shutdown mode selected by $605E_h$ ($03C1_h$); otherwise: shut down the outputs according to the slow curve stop.

2- The state machine switches to the fault state (refer to "4.1 CiA402 state machine"), if $605E_h$ ($03C1_h$) >= 0, then the fault shutdown mode is selected according to $605E_h$ ($03C1_h$); otherwise: the output is shut down according to the fast curve stop.

3- Pressing the quick curve suspends the locking axis without switching to the fault state and is not affected by object $605E_h$ ($03C1_h$).

4- Warning and does not change the current operating state, does not switch to faulty state and is not affected by object $605E_h$ ($03C1_h$).

5- Ignored and logged in 1003_h (0002_h ~0020_h), not switched to the fault state and not affected by object $605E_h$ ($03C1_h$).

6- Ignored but not logged in 1003_h (0002_h ~0020_h), not switched to fault state and not affected by object $605E_h$ ($03C1_h$).

Alarm Code Configuration

Alarm codes that can be freely configured by the user are exemplified in fault code group $200E_h$ ($025B_h$ ~$027B_h$) as follows:

Table 8- 2

Fault Code Group $200E_h$ ($025B_h$ ~027B)$_h$	**hidden meaning**	**Default Failure Action**
0x4012311	Motor overload	No
0x3002312	motor stalling	No
0x13210	Power overvoltage	Yes
0x1013220	Power supply undervoltage	Yes
0x14210	Over Temperature Alarm	Yes
0x14220	Low temperature alarm	Yes
0x16320	Parameter setting error	Yes
0x17310	speeding	Yes
0x17501	Illegal Function Codes in Modbus	No

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Fault Code Group 200Eh (025Bh ~027B)h	hidden meaning	Default Failure Action
	Communication	
0x17502	Illegal addresses in Modbus communication	No
0x17503	Illegal data values in Modbus communication	No
0x17505	Confirmation in Modbus communication	No
0x17506	Slave device busy in Modbus communication	No
0x1750C	Synchronization message in Modbus communication requesting data data is greater than the total mapped data	No
0x1750D	The number of synchronization messages requesting data in Modbus communication does not correspond to the mapping data equally.	No
0x1750E	Wrong node address for unicast message under synchronization function in Modbus communication	No
0x8610	Home return timeout	Yes
0x8611	The location is terrible.	Yes
0x3018613	Software limit error	No
0x3018614	Limit switch error	No
0x3285	Output phase failure	Yes
0x4018615	Errors in curve planning calculations	No
0x18616	Target position overflow	Yes
0x5018617	Curve planning parameters too small	No

Note: The "Yes (No)" for the fault action in the above list is an indication of whether or not the motor entered a fault state after this fault was generated.

Alarm Code Configuration Examples

Power supply undervoltage setting

200Eh :04h (0261h) (power supply undervoltage) is set to 0x1003220, when the motor is running, the power supply voltage is lower than the detection threshold (default 24V), the motor is shut down according to the shutdown mode set by 605Eh (03C1h) (refer to the order of the corresponding bits, the fault response code corresponds to 1); when the power supply voltage is restored to normal, it is necessary to operate the motor by manual reset When the power supply voltage returns to normal, the motor needs to be reset manually before it can be operated normally (refer to the order of the corresponding bits, the self-reset permission corresponds to 0).

8.3 Fault Reset

After the integrated fault recovery, the fault codes that are not allowed by the self-reset need to be

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operated by manual reset operation before the motor can be operated for normal running operation. There are 3 ways of manual reset operation as follows:

- After the motor is de-enabled, the fault state is cleared by sending the control word 0x80 (control word bit7 rising edge active)
- Through the entity input terminal 2003_h (00D5h ~$00E6_h$) one of the functions is configured as the No. 2 function code (alarm reset), and set the corresponding logic selection, according to the logic selection of its setup terminal to make it effective can be reset!
- Configure one of the functions of virtual input terminal 2017_h (0326_h ~0346_h) as function code No. 2 (alarm reset) and set the corresponding logic selection, and then reset the alarm by making it valid according to the logic selection of the setup terminal.

8.4 Fault Detection Instructions

- Drive Failure Detection

  This fault is triggered by the drive chip fault signal interrupt response, triggering is reported as a drive fault.

- I^2 T overload detection

  When the motor is running above the rated current, the fault detection will calculate the time integral of I^2 . When the integral value is greater than the product of the set maximum current 2000_h : $0A_h$ (0066_h) and the maximum current duration 2000_h : $0B_h$ (0067_h), it will report an overload warning and limit the torque output to the rated value, and then unrestrict it after 30s.

- Blocking fault detection

  When the motor is running at a speed lower than 5 rpm and above the rated current, the fault detection will calculate the time integral of I2 and report a blocking fault and stop the output when the integral value is greater than the product of the set maximum current 2000_h : $0A_h$ (0066_h) and the maximum current duration 2000_h : $0B_h$ (0067_h).

- High and low temperature fault detection

  When the drive temperature is greater than 105 ° C, report high temperature fault, below 70 ° C, the fault is lifted; when the drive temperature is lower than -25 ° C, report low temperature fault, higher than -20 ° C, the fault is lifted.

- Over-voltage and under-voltage fault detection

  An overvoltage fault is reported when the motor's supply voltage is higher than the voltage set at 2001_h : $0E_h$ (0092_h), and an undervoltage fault is reported when the motor's supply voltage is lower than the voltage set at 2001_h : 10_h (0094_h).

- Parameter setting fault detection

  Parameter Setting Fault Detection Errors in the setting of the multifunction terminals, including the same function number being set for different terminals (physical and virtual) or the 14, 15 and 31 function numbers not being set for the CiA402 home return mode.

- Over-speed fault detection

  When the motor speed is higher than the speed set by $200A_h$: 06_h ($01BC_h$) and maintained for 10ms, an overspeed fault is reported, and lower than that, the fault state is released.

- Illegal address fault detection in Modbus communication

  This error is identified by the slave protocol error flag bit under Modbus communication.

- Illegal data value fault detection in Modbus communication

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This error is identified by the slave protocol error flag bit under Modbus communication.

- Illegal Function Code Fault Detection in Modbus Communication
  This error is identified by the slave protocol error flag bit under Modbus communication.
- Fault Detection of Unicast Message Node Address Error under Synchronization Function in Modbus Communication
  This error is identified by the slave protocol error flag bit under Modbus communication.
- Slave device busy fault detection in Modbus communication
  This error is identified by the slave protocol error flag bit under Modbus communication.
- Synchronization message in Modbus communication requesting data data greater than mapped total data fault detection
  This error is identified by the slave protocol error flag bit under Modbus communication.
- Fault detection of unequal mapping data corresponding to the number of data requested in synchronization messages in Modbus communication
  This error is identified by the slave protocol error flag bit under Modbus communication.
- Fault Detection of Unicast Message Node Address Error under Synchronization Function in Modbus Communication
  This error is identified by the slave protocol error flag bit under Modbus communication.
- Hardware initialization failure detection

If the hardware (driver, encoder) is not initialized successfully when the motor is powered up, a hardware initialization fault is reported.

- Flash Fault Detection
  When the motor's flash is operated, a flash failure is reported when the saved parameters or power-on reading of the stored parameters do not pass the test.
- Home Return Timeout Fault Detection
  When the motor is in CiA402 home return mode enable, if the home return action cannot be completed within the limited time 2005_h : $1C_h$ ($012E_h$), the home return timeout fault is reported.
- Tracking Fault Detection
  The motor reports a tracking fault when the position deviation exceeds ±6065_h ($03CA_h$) and the time reaches 6066_h ($03CC_h$) (ms) under position mode control.
- Software overrun fault detection
  When the motor is in position control mode, it is judged according to the software limit value (user unit) set by $607D_h$ ($03EF_h$ ~$03F1_h$), and a software overrun fault is reported if the actual position exceeds the set value.
- Overrun Switch Fault Detection
  According to the function numbers 14 and 15 set in the multi-function terminal, it detects whether the positive or negative limit switch is triggered or not, and if it is triggered, it reports an over-limit switch fault.
- Target position overflow fault detection
  The motor is in position control mode, the given position $607A_h$ ($03E7_h$) after the gear ratio 6091_h ($040E_h$ ~0410_h) exceeds the data range of data type int32 (-2147483648~+2147483647) then the target position overflow fault is reported.
- Fault detection of too small curve planning parameters

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If the motor is in position control mode and the given impact or acceleration fails to realize S-curve in the given displacement interval, the curve planning parameter is reported to be too small.

- Flash initialization fault (0x5541)

  Controller is not saving Nimotion zone parameters, contact factory personnel.
- Commands that cannot be executed

  When the motor is in the enable state, or when the supply voltage is less than 80% of the rated voltage 2000_h :07_h (0063_h), the motor reports a fault of not being able to execute commands when operating the save parameter and the restore parameter.
- Encoder low voltage fault detection

  When the servo motor is powered on, the encoder battery voltage is detected to be ≤2.4V, and a fault 0x1007301 is reported.
- Encoder low voltage alarm detection

  When the servo motor is powered on, the encoder battery voltage is detected to be between 2.4V and 2.6V, and a alarm 0x4017302 is reported.
- Encoder communication fault detection

  The servo motor fails to communicate with the encoder and reports a 0x7303 fault.
- Encoder multi-turn counting error detection

  If the encoder is not calibrated or the calibration fails, a 0x7304 fault is reported.
- Internal fault of the encoder detection

  When the power supply power is lost, the battery disconnects or other situations cause the encoder to abnormally restart, reporting the 0x7309 fault. When this fault is reported, connect the battery, restart the power supply of the servo motor, and reset the position to zero.
- STO_1 failed to close detection

  STO_1 The input signal is high, the power drive power supply is disconnected, and the 0XFF05 fault is reported. The STO_2 input signal is high, the power drive power supply is disconnected, and a 0XFF07 fault is reported.
- STO_1 failed to enable detection

  The input signal of STO_1 is low, the power drive power supply is connected, and the 0XFF06 fault is reported. The input signal of STO_2 is low, the power drive power supply is connected, and the 0XFF08 fault is reported.
- STO input abnormal fault detection

  If either of the input signals STO_1 or STO_2 becomes high and the other remains low for more than 6ms, a 0xFF09 fault is reported.
- STO enabled state detection

  When at least one input signal of STO_1 or STO_2 drops, the STO enable state is entered. The power drive power supply and drive signal are cut off, and the servo enable command becomes invalid, reporting 0xFF0A.
- Non-safe state

  Each time the power is turned on or the STO is restored, the motor and encoder are not inspected completely or fail the inspection.

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9 NiMotion Protocol

The NiMotion protocol optimizes the Modbus protocol and creates two new message formats (process data message and synchronization message) on top of Modbus. Through these two telegrams can achieve multi-command control, greatly reduce the data processing time, improve the coordination of multi-motor control, and further enable the motor to achieve two-two synchronization function. The process data message is used for the free combination of multiple parameters into a frame message to send and receive processing at the same time, and the synchronization message is used to control the timing of the effective command.

R_PDO: Receive process data;

T_PDO: Send process data.

9.1 Master Slave Timing Diagram

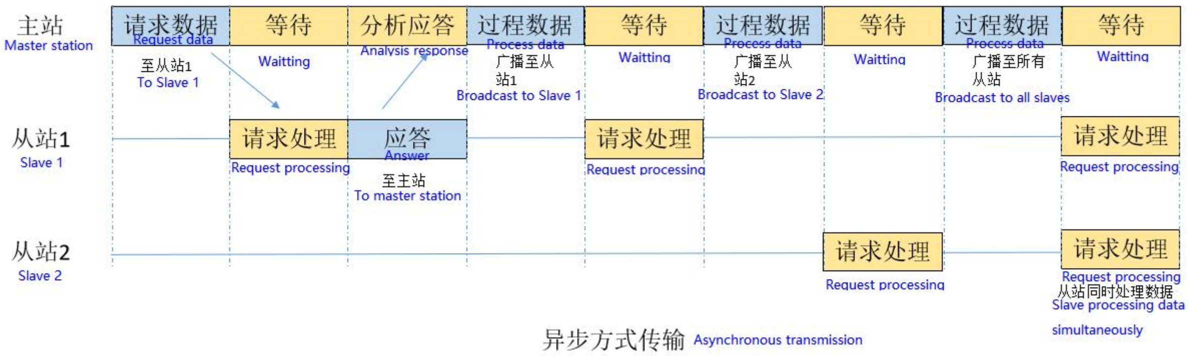

Figure 9- 1

In the asynchronous transmission process, the unicast process data for slave 1 sends an answer to the master after being processed by slave 1, and the master receives an error-free answer and sends the broadcast process data for slave 1, which is only processed by slave 1, but there is no answer; the master sends the broadcast process data for slave 2, which is processed by slave 2, but there is no answer; and the master sends the broadcast process data for all the slaves The master sends broadcast process data for all the slaves, and all the slaves process it, but there is no answer.

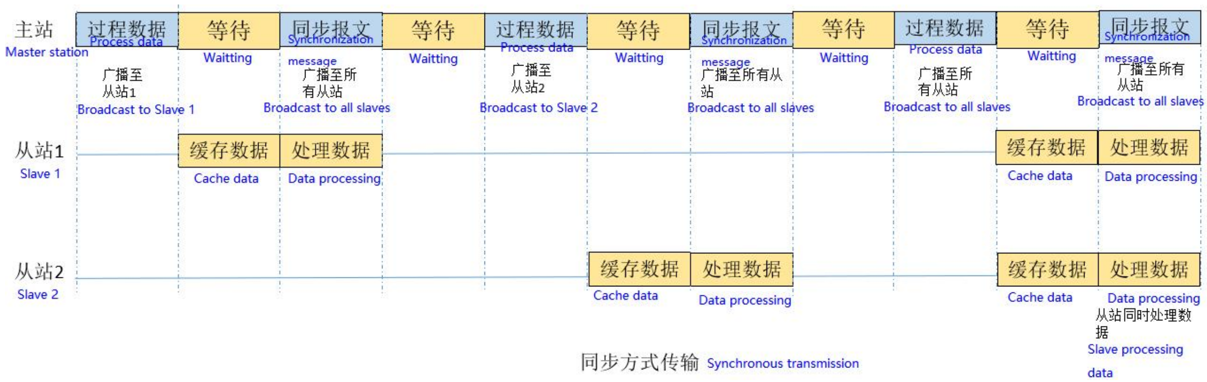

Figure 9- 2

During synchronous transmission, the master sends broadcast process data for slave 1, slave 1 receives the message and just caches the data and does not process it, the master broadcasts and sends the synchronization message again, and slave 1 starts processing the data; the master sends

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broadcast process data for slave 2, slave 2 receives the message and just caches the data and does not process it, the master broadcasts and sends the synchronization message again, and slave 2 starts processing the data; the master sends broadcast process data for all slaves, slave 1 and slave 2 receive the message and both cache the data and do not process it, the master broadcasts and sends the synchronization message again, and both slave 1 and slave 2 start processing the data. The master sends broadcast process data for all slaves, slave 1 and slave 2 receive the message and just cache the data and do not process it, the master broadcasts a synchronization message again and slave 1 and slave 2 start to process the data at the same time.

9.2 Message Format

9.2.1 synchronization message

A synchronization message is a broadcast message used to determine when process data is updated to a register in synchronization mode by redefining the meaning of the register address.

The function of the synchronization telegram is to update the data stored in the buffer to the registers in the synchronization mode and thus control the slave changes.

Synchronous broadcast messages are specified as follows:

Table 9- 1 Master synchronization request messages

Serial Number	Field Name	Byte Length (bytes)	Element	Note
1	address field	1	0x00	broadcast mode
2	function code	1	0x10	Write multiple registers
3	Register Address High Byte	1	0x80	Synchronization message identifier 0x8000
4	Register Address Low Byte	1	0x00	
5	Number of registers high byte	1	0x00	1 register
6	Number of registers low byte	1	0x01	
7	byte count	1	0x02	2 bytes of data portion
8	Register Value High Byte	1	0x66	Synchronized message data value flag Fixed to 0x6688
9	Register Value Low Byte	1	0x88	
10	CRC16 Low Bit	1	Calculate CRC low byte	16-bit CRC checksum
11	CRC16 High	1	Calculate CRC high byte	

Synchronization message is to write data to the slave, considering the process data message is to write multiple registers, so in order to reduce the change of the existing program, so the uniform selection of 0x10 function code.

In order to develop a specific synchronization message, the register address is changed to the

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identification of the synchronization message, and the highest bit is set to 1, 0x8000, considering the need to distinguish it from the existing register address.

The register value acts as a synchronization flag, so the fixed data is 0x6688.

9.2.2 Process Data Write Operations

9.2.2.1 request message

Support modbus protocol message format

Read content, length, set according to registers (0x6000~0x6021).

Table 9- 2 Process data requests

Serial Number	Field Name	Byte Length (bytes)	Element	Note
1	address field	1	0x00	Broadcast/unicast mode
2	function code	1	0x10	Write multiple registers
3	Register Address High	1	0x83	message mode
4	Register Address Low	1	0x00	
5	Number of registers high byte	1	0x00	Number of registers (1~33)
6	Number of registers low byte	1	0x05	
7	byte count	1	0x0A	Number of registers*2 (2~66)
8	Register Value High Byte	1	0x00	Motor address (0~247) 0 for sending to all motors
9	Register Value Low Byte	1	0x01	
10	Register Value High Byte	1	0x00	The value of the register (set up according to actual needs)
11	Register Value Low Byte	1	0x07	
12	Register Value High Byte	1	0x00	
13	Register Value Low Byte	1	0x01	
14	Register Value High Byte	1	0x00	
15	Register Value Low Byte	1	0x00	
16	Register Value High Byte	1	0x03	

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Serial Number	Field Name	Byte Length (bytes)	Element	Note
17	Register Value Low Byte	1	0x84	
18	CRC16 Low Bit	1	Calculate CRC low byte	16-bit CRC checksum
19	CRC16 High	1	Calculate CRC high byte	

The process data is written to the slave for multiple registers, so the 0x10 function code is used.

In order to develop a specific process data message, the register address is changed to the identification of the process data message. Considering the need to distinguish it from the existing register address, the highest bit is set to 1. Referring to the CAN customized process data format, the process data identification is +0x300, i.e. 0x8300.

The mapping has a maximum of 16 entries, and each entry has a maximum of two registers, so the mapping has a maximum of 32 registers, plus a motor address accounts for one register, so the number of registers ranges from 1 to 33.

Each register occupies two byte counts, so the range of byte counts is 2 to 66.

The node address occupies one register, and there can only be a maximum of 247 slaves on the modbus bus. So the range is 1~247. if it is 0, it means send to all slaves.

9.2.2.2 Reply Message

Refer to the modbus protocol 0x10 function code response message.

9.2.3 Process Data Read Operation

9.2.3.1 Request Message

The process data read operation can only be used in asynchronous mode and supports the modbus protocol message format.

Read content, length, set according to registers (0x6000~0x6021).

Table 9- 3 Process data read request messages

Serial Number	Field Name	Byte Length (bytes)	Element	Note
1	address field	1	0x01	node address
2	function code	1	0x04	Read multiple registers
3	Register Address High	1	0x83	Register Address (Buffer Address)
4	Register Address Low	1	0x80	
5	Number of registers high byte	1	0x00	Number of registers (1~32)
6	Number of	1	0x02	

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Serial Number	Field Name	Byte Length (bytes)	Element	Note
	registers low byte			
13	CRC16 Low Bit	1	Calculate CRC low byte	16-bit CRC checksum
14	CRC16 High	1	Calculate CRC high byte	

Function code 0X10 is to write the register, here is to read the value of the register, so 0x10 function code is no longer appropriate, choose 0x04 function code

The process data is read from the cache, so the address is also the address of the cache.

There are up to 16 mapped entries in the cache, and each entry has at most two registers, so the number of registers ranges from 1 to 32.

9.2.3.2 Reply Message

Refer to the modbus protocol 0x04 function code response message.

9.3 Exception Handling

Refer to Chapter 8, Fault Management.

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9.4 Example

1. Set the Slave Buffer Mapping Entry Count Register (0x5000) to determine the number of entries as required.
2. Depending on the number of entries, the setting starts from the R_PDO mapped 1 register and can be set up to a maximum of mapped 12 registers, with the register high byte determining the start address of the register to be mapped to, and the register low byte determining the number of registers to be mapped.
3. Set the Slave Buffer Mapping Entry Number Register (0x6000) to determine the number of entries as required.
4. Depending on the number of entries, the setting starts from T_PDO mapping 1 register and can be set up to a maximum of mapping 12 registers, with the register high byte determining the start address of the register to be mapped to and the register low byte determining the number of registers to be mapped.
5. Set Synchronization Mode Enable $200C_h$: 18_h (0246_h) to enable (0x01).
6. The master sends a process data message with register address 0x8300 and the slave executes the function according to the function code.
7. The master sends a synchronization message with register address 0x8000 and data value 0x6688, and the slave executes the function code function. The data update is triggered and the motor status is changed accordingly.

Realize the process:

1. The number of RPDO mapping entries for motor #1 is 2.

Table 9- 4

Slave Address	Function Code	Register Address	Number of Registers	Byte Count	Register Value	CRC Value
01	0x10	50 00	00 02	04	00 00 00 02	-

2. No. 1 motor RPDO mapping control word 6040_h : 0_h (0380_h), target position $607A_h$: 0_h ($03E7_h$).

Table 9- 5

Slave Address	Function Code	Register Address	Number of Registers	Byte Count	Register Value	CRC Value
01	0x10	50 02	00 04	08	03 80 00 01 03 E7 00 02	-

3. No. 1 motor on synchronization mode $200C_h$: 18_h (0246_h).

Table 9- 6

Slave Address	Function Code	Register Address	Register Value	CRC Value
01	0x06	02 46	00 01	-

4. Motor No. 1 saves the parameters and restarts with power off.

Table 9- 7

Slave Address	Function Code	Register Address	Number of Registers	Byte Count	Register Value	CRC Value

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01	0x10	00 26	00 02	04	65 76 61 73	-

5. The master broadcast sends process data (control word 6, target position 1000) to motor No. 1.

Table 9- 8

Slave Address	Function Code	Register Address	Number of Registers	Byte Count	Register Value	CRC Value
00	0x10	83 00	00 04	08	00 01 00 06 00 00 03 E8	-

6. The master broadcast sends a synchronization message to update the parameters.

Table 9- 9

Slave Address	Function Code	Register Address	Number of Registers	Byte Count	Register Value	CRC Value
00	0x10	80 00	00 01	02	66 88	-

9.5 Mapping Parameter Settings

This mapping is made for process data multi-command control, each mapping entry occupies two registers, the high register is used to store the register first address of the parameter that the user wants to map, and the low register is used to store the number of registers that the parameter occupies in the parameter table. Addresses 0x5000 and 0x6000 represent the number of mapping entries enabled by R_PDO and T_PDO, respectively, and the maximum number of mappable entries is 12.

Example: the value of R_PDO map 1 is 0x02310001

The high 16 bits, 0x0231, refer to the parameter mapped to register address 0x0231: communication baud rate.

The low 16 bits 0x0001 means that the parameters of this map occupy a register.

Table 9- 10

Modbus address	Name (of a thing)	Description	Number of Registers	Data Range
5000	R_PDO Number of mapping entries	Number of entries in the process data message mapping registers	2	0~0x0C
5002	R_PDO mapping parameter 1	High register: mapped register start address, Low register: number of registers	2	0~4294967295
5004	R_PDO mapping parameter 2	High register: mapped register start address , Low register: number of registers	2	0~4294967295
5006	R_PDO mapping parameter 3	High register: mapped register start address , Low register: number of registers	2	0~4294967295
5008	R_PDO mapping parameter 4	High register: mapped register start address , Low register: number of registers	2	0~4294967295

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Modbus address	Name (of a thing)	Description	Number of Registers	Data Range
500A	R_PDO mapping parameter 5	High register: mapped register start address , Low register: number of registers	2	0~4294967295
5006C	R_PDO mapping parameter 6	High register: mapped register start address, Low register: number of registers	2	0~4294967295
500E	R_PDO mapping parameter 7	High register: mapped register start address , Low register: number of registers	2	0~4294967295
5010	R_PDO mapping parameter 8	High register: mapped register start address , Low register: number of registers	2	0~4294967295
5012	R_PDO mapping parameter 9	High register: mapped register start address , Low register: number of registers	2	0~4294967295
5014	R_PDO mapping parameter 10	High register: mapped register start address , Low register: number of registers	2	0~4294967295
5016	R_PDO mapping parameter 11	High register: mapped register start address , Low register: number of registers	2	0~4294967295
5018	R_PDO mapping parameter 12	High register: mapped register start address, Low register: number of registers	2	0~4294967295
6000	Number of T_PDO mapping entries	Number of entries in the process data message mapping registers	2	0~0x0C
6002	T_PDO mapping parameter 1	High register: mapped register start address , Low register: number of registers	2	0~4294967295
6004	T_PDO mapping parameter 2	High register: mapped register start address , Low register: number of registers	2	0~4294967295
6006	T_PDO mapping parameter 3	High register: mapped register start address , Low register: number of registers	2	0~4294967295
6008	T_PDO mapping parameter 4	High register: mapped register start address , Low register: number of registers	2	0~4294967295
600A	T_PDO mapping parameter 5	High register: mapped register start address , Low register: number of	2	0~4294967295

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Modbus address	Name (of a thing)	Description	Number of Registers	Data Range
		registers		
600C	T_PDO mapping parameter 6	High register: mapped register start address , Low register: number of registers	2	0~4294967295
600E	T_PDO mapping parameter 7	High register: mapped register start address , Low register: number of registers	2	0~4294967295
6010	T_PDO mapping parameter 8	High register: mapped register start address , Low register: number of registers	2	0~4294967295
6012	T_PDO mapping parameter 9	High register: mapped register start address , Low register: number of registers	2	0~4294967295
6014	T_PDO mapping parameter 10	High register: mapped register start address , Low register: number of registers	2	0~4294967295
6016	T_PDO mapping parameter 11	High register: mapped register start address , Low register: number of registers	2	0~4294967295
6018	T_PDO mapping parameter 12	High register: mapped register start address , Low register: number of registers	2	0~4294967295

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10 Object Dictionary

The object dictionary is an important part of the device specification, it is an ordered set of parameters and variables that contains all the parameters of the device description and the device network state. Each object contains an index, subindex, name, description, data type, data range, accessibility, whether it can be mapped, factory settings, units, and how it takes effect.

interpretation of nouns

"Index": specifies the position of objects of the same class in the object dictionary, expressed in hexadecimal
"Sub-index": under the same index, which contains multiple objects, the bias of each object under the class
"Data types and ranges": please refer to the following table for details

Table 10- 1

Data Type	Data Range	Data Length
Int8	-128~127	1 byte
Int16	-32768~32767	2 bytes
Int32	-2147483648~2147483647	4 bytes
Uint8	0~255	1 byte
Uint16	0~65535	2 bytes
Uint32	0~4294967295	4 bytes
String	ASCII	~

"Accessibility": please refer to the following table for details

Table 10- 2

Accessibility	Clarification
RW	readable
WO	write only
RO	read-only (computing)

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"Can it be mapped": see the table below for details

Table 10- 3

Mappable	Clarification
NO	Not mappable in PDO
RPDO	Can be used as an RPDO
TPDO	Can be used as a TPDO

"Mode of entry into force": please refer to the following table for details

Table 10- 4

Mode of entry into force	Clarification
immediate effect	The set values take effect immediately after the parameter setting is completed.
Suspension in effect	After the parameter setting is completed, wait until the drive is not in the operation state and the setting value takes effect.
Re-energized.	After the parameter setting is completed, turn on the power of the drive again and the set value takes effect

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10.1 Communication Parameters 1000h Group

The 1000h object group contains the parameters required for Modbus communication.

Table 10- 5

Index	Sub-index	Name	Description	Data type	Data range	Access-ibility	Mapp-ability	Factory setting	Unit	Activate method
1001$_h$	-	00$_h$	error register	Uint8	-	RO	NO	0	VAR	-
1003$_h$	00$_h$	01$_h$	Number of alarms present in the current device	uint8	-	RO	NO	-	VAR	-
	01$_h$	02$_h$	History Alarm Cache	uint32	-	RO	NO	-	VAR	-
	02$_h$	04$_h$		uint32	-	RO	NO	-	VAR	-
	03$_h$	06$_h$		uint32	-	RO	NO	-	VAR	-
	04$_h$	08$_h$		uint32	-	RO	NO	-	VAR	-
	05$_h$	0A$_h$		uint32	-	RO	NO	-	VAR	-
	06$_h$	0C$_h$		uint32	-	RO	NO	-	VAR	-
	07$_h$	0E$_h$		uint32	-	RO	NO	-	VAR	-
	08$_h$	10$_h$		uint32	-	RO	NO	-	VAR	-
	09$_h$	12$_h$		uint32	-	RO	NO	-	VAR	-
	0A$_h$	14$_h$		uint32	-	RO	NO	-	VAR	-
	0B$_h$	16$_h$		uint32	-	RO	NO	-	VAR	-
	0C$_h$	18$_h$		uint32	-	RO	NO	-	VAR	-
	0D$_h$	1A$_h$		uint32	-	RO	NO	-	VAR	-
	0E$_h$	1C$_h$		uint32	-	RO	NO	-	VAR	-
	0F$_h$	1E$_h$		uint32	-	RO	NO	-	VAR	-
	10$_h$	20$_h$		uint32	-	RO	NO	-	VAR	-
1010$_h$	01$_h$	26$_h$	Save all current parameters	uint32	-	RW	NO	1	VAR	-

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Index	Sub-index	Name	Description	Data type	Data range	Access-ibility	Mapp-ability	Factory setting	Unit	Activate method
1011_h	01_h	$3C_h$	Read all parameters	uint32	-	RW	NO	1	VAR	-
1017_h	-	52_h	Producer heartbeat time	Uint16	-	RW	NO	0	ms	-
1018_h	03_h	53_h	software version number	uint32	-	RO	NO	-	-	-
	04_h	55_h	Product Serial Number	uint32	-	RO	NO	-	-	-

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10.2 Manufacturer Defined Parameters 2000h Group Description

Table 10- 6

Index	Sub-index	Name	Description	Data type	Data range	Access-ibility	Mapp-ability	Factory setting	Unit	Activate method
2000h (motor parameters)	06h	62h	Bus Encoder Type	uint16	0~1	RW	NO	1	-	Suspension in effect
	07h	63h	rated voltage	uint16	24~48	RW	NO	48	1V	Suspension in effect
	08h	64h	rating	uint16	-	RW	NO	40	0.01KW	Suspension in effect
	09h	65h	rated current	uint16	-	RW	NO	1250	0.01A	Suspension in effect
	0Ah	66h	Maximum current	uint16	-	RW	NO	2250	0.01A	Suspension in effect
	0Bh	67h	Maximum current duration	uint16	-	RW	NO	3000	0.1s	Suspension in effect
	0Ch	68h	Rated torque	uint16	-	RW	NO	127	0.01Nm	Suspension in effect
	0Dh	69h	Maximum torque	uint16	-	RW	NO	2000	0.01Nm	Suspension in effect
	0Eh	6Ah	rated speed	uint16	-	RW	NO	3000	1rpm	Suspension in effect
	0Fh	6Bh	Maximum speed	uint16	-	RW	NO	6000	1rpm	Suspension in effect
	10h	6Ch	Moment of inertia Jm	uint16	-	RW	NO	30	0.01kgcm^2	Suspension in effect
	11h	6Dh	Permanent magnet	uint16	-	RW	NO	4	1	Suspension

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Index	Sub-index	Name	Description	Data type	Data range	Access-ibility	Mapp-ability	Factory setting	Unit	Activate method
			synchronous motor pole pair number							in effect
	12_h	$6E_h$	Stator resistance	uint16	-	RW	NO	115	0.001Ω	Suspension in effect
	13_h	$6F_h$	Stator inductance Lq	uint16	-	RW	NO	25	0.01mH	Suspension in effect
	14_h	70_h	Stator inductance Ld	uint16	-	RW	NO	25	0.01mH	Suspension in effect
	16_h	72_h	Torque coefficient Kt	uint16	-	RW	NO	11	0.01Nm/Arms	Suspension in effect
	19_h	75_h	Encoder absolute digits	uint32	-	RW	NO	14	1bit	Suspension in effect
	$1A_h$	77_h	Number of encoder poles	uint16	-	RW	NO	1	1	Suspension in effect
	$1C_h$	$7A_h$	Position sensor selection	uint16	-	RW	NO	2	-	Suspension in effect
	$1D_h$	$7B_h$	Z signal corresponds to the electrical angle	uint16	-	RW	NO	0	1/4096	Suspension in effect
	$1E_h$	$7C_h$	Rising edge of U-phase corresponds to the diagonal angle	int16	-	RW	NO	0	1	Suspension in effect
2001_h (drive parameters)	07_h	$8B_h$	Rated output current	Uint16	0~65535	RO	NO	1200	0.01A	Suspension in effect
	08_h	$8C_h$	Maximum Output Current	uint16	0~65535	RO	NO	1500	0.01A	Suspension in effect
	$0D_h$	91_h	Switching Dead Time	uint16	0~65535	RW	NO	100	us	Suspension

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